MECHANICAL ENGINEERING PORTFOLIO

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4.00
M.S. GPA

3.65
B.S. GPA

M.S. Mechanical Engineering • University of Florida • 4.00 GPA

3	4.00	10+	Cum Laude
FEATURED PROJECTS	GRADUATE GPA	TOOLS & SOFTWARE	B.S. DISTINCTION

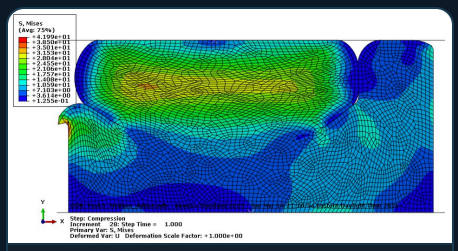
PROJECT 01

Digest-A-Bull

Senior Capstone, 2024-2025

Biomimetic cow rumen simulator for agricultural research. Led mechanical design of 3 subsystems: heating, mixing, and vessel/lid. Reduced setup time from multiple days to 79 minutes.

SolidWorks | PID Control | Thermal Modeling | ESP32



PROJECT 02

Engine Gasket Nonlinear FEA

Graduate, EGM 6352, 2026

Nonlinear FEA of a rubber gasket in Abaqus. Calibrated Ogden hyperelastic model (R^2=0.9629) and redesigned geometry to eliminate a 2.22 mm sealing gap.

Abaqus | MATLAB | Nonlinear FEA | Hyperelastic

PROJECT 03

No-Slip Walker

ME Design 2, EML 4501, 2024

Active fall-prevention walker for elderly users rated to 300 lb. Defined load cases, selected 6061-T6 aluminum, and ran ANSYS FEA on 4 key structural components.

ANSYS R22.2 | GD&T | Fusion 360 | Structural FEA

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B.S. ME, Cum Laude | M.S. ME, In Progress | University of Florida

Digest-A-Bull

Biomimetic Cow Rumen Digestion Simulator

Senior Capstone | 10-Member Team
2024-2025

Fig. 1 Front view CAD render — vessel, lid, mixing shaft, and electronics base

Fig. 2 Isometric render with effluent collection bottle and valve assembly

PROJECT OVERVIEW

Digest-A-Bull is a biomimetic cow rumen simulator designed to replicate the anaerobic fermentation environment of a bovine rumen for agricultural research at UF. I led mechanical design of three subsystems: heating, mixing, and vessel/lid assembly.

KEY OUTCOMES

- Setup time reduced from multiple days to **79 minutes**
- Research throughput increased by **25%**
- Temperature held within **±0.1°C** across all 5 setpoints
- IPX-5 sealed, pressure-safe, IoT-connected operation

VOLUME	TEMP RANGE	ACCURACY	SEALING
1.8 L	30-50°C	±0.1°C	IPX-5

TOOLS & METHODS

SolidWorks · MATLAB · ESP32 · PID Control · Thermal Modeling · SolidWorks FEA · Pugh Chart · 316L Stainless Steel · Borosilicate Glass · PT100 RTD

Design process: 10 concepts per member, Pugh chart vs. legacy baseline, weighted scoring (10 to 4 to 1), nine-box risk prioritization, hand calculations validated against SolidWorks FEA, two-protocol testing.

SUBSYSTEM 1 — HEATING SYSTEM

The legacy heat wrap was replaced with dual stainless steel immersion heater rods for direct fluid contact. I built a thermal resistance network (Fig. 3) capturing conduction through the borosilicate glass vessel wall, 316L steel lid, and silicone insulation pad, plus natural convection losses to ambient air.

Thermal Analysis and Control

- Energy balance predicted 23-min heat-up for 1.8 L from 22°C to 50°C
- Validated in hardware: actual heat-up 13 to 17 min, exceeding model prediction
- PID controller on ESP32 with PT100 RTD feedback
- Brushless DC gearmotor selected to eliminate EMI interference with RTD signal

VALIDATION DATA (5 SETPOINTS)

- 30°C: avg 29.95°C — error 0.04°C
- 35°C: avg 34.90°C — error 0.10°C
- 40°C: avg 39.93°C — error 0.075°C
- 45°C: avg 44.94°C — error 0.06°C
- 50°C: avg 49.93°C — error 0.06°C

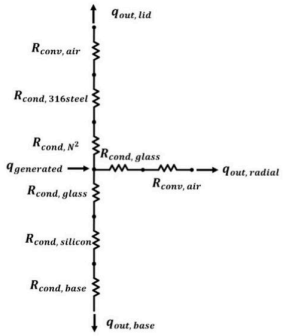

Fig. 3 Thermal resistance network: conduction through lid/glass/silicone/base; convection losses radial and through lid

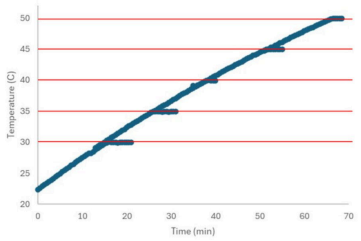

Fig. 4 Temperature vs. time for all 5 setpoints. All within ±0.1°C steady state.

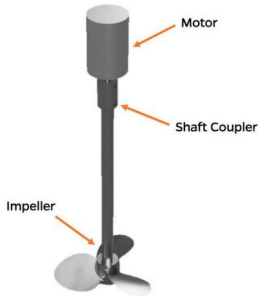

Fig. 5 Mixing assembly: BLDC gearmotor, shaft coupler, and 76.2 mm pitched-blade impeller (Caframo)

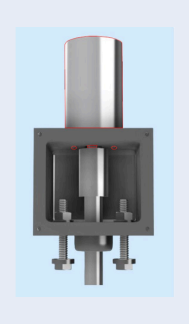

Fig. 6 Box motor mount (CNC-machinable). Redesigned from cylindrical form for standard 3-axis mill production.

Fig. 7 Full system isometric — integrated lid, mixing, and vessel subsystems

SUBSYSTEM 2 — MIXING SYSTEM

The mixing system was designed to maintain solid suspension in approximately 3,000 cP simulated rumen fluid. I selected a 76.2 mm pitched-blade propeller (Caframo) and performed engineering calculations to size the motor and validate shaft integrity.

Engineering Calculations

- Reynolds number calculated for impeller in high-viscosity fluid to confirm mixing regime
- Power number correlation used to predict required motor power output
- Torque and shaft stress calculations to verify structural adequacy
- Proportional speed control at ±1.5 RPM precision

Motor and EMI Mitigation

Replaced brushed DC motor with a 12V, 159 RPM brushless DC gearmotor to eliminate electromagnetic interference with the PT100 RTD sensor. Brushed commutation noise corrupts RTD readings in close-proximity installations.

MANUFACTURABILITY REDESIGN

The motor mount was redesigned from cylindrical to box geometry (Fig. 6). The cylindrical design required specialized turning operations; the box form is machinable on a standard 3-axis CNC mill from 316L stainless steel stock, reducing cost and enabling in-house production.

IMPELLER	MOTOR	SPEED	VISCOSITY
76.2 mm	**12V BLDC**	**±1.5 RPM**	**~3000 cP**
Caframo pitched-blade	159 RPM gearmotor	Proportional	Simulated rumen

SUBSYSTEM 3 — VESSEL AND LID ASSEMBLY

Fig. 8 Lid underside — heater rod ports, O-ring groove, mounting features

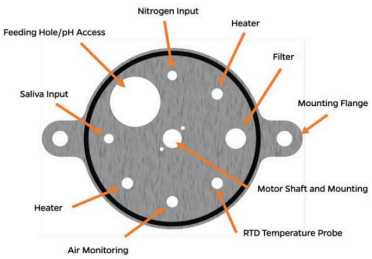

Fig. 9 Lid port layout — all 9 ports labeled to eliminate subsystem interference

Port Layout and Sealing

- Port layout redesigned to prevent interference between feeding tube, heaters, sensors, and motor shaft
- O-ring groove per ASME standards; lid deflection analyzed at operating pressure
- Pressure-tight feeding tube eliminates need to depressurize vessel between feeds
- Outflow port angled 45° downward to reduce clogging

Safety Systems

- MQ-4 methane sensor for fermentation gas detection
- Pressure relief valve and ±2 kPa differential pressure sensor
- pH probe port for real-time fermentation monitoring
- Overflow routing for over-pressure protection
- ESP32 with micro-SD logging and IoT cloud

DESIGN PROCESS

Concept Generation and Selection

- 10 concepts per team member — 100 total ideas across the 10-member team
- Pugh chart compared all concepts against legacy system baseline
- Weighted scoring (10 to 4 to 1) narrowed design space from concept to prototype
- Nine-box risk prioritization matrix identified and ranked top failure modes

Analysis and Validation

- Hand calculations for all subsystem sizing, validated against SolidWorks FEA
- Two-protocol test plan: benchtop functional tests and full integrated system tests
- Temperature setpoints verified across 5 independent targets
- Pressure sealing validated at operating pressure with margin

MY ROLE

Within the 10-member team, I owned full mechanical design of the three subsystems: all hand calculations, SolidWorks CAD modeling, SolidWorks FEA validation, material selection, and DFM review. I also contributed to integration testing and documentation.

ENGINEERING VALUE DELIVERED

Switching to immersion heaters cut heat-up time to 13-17 min vs. the 23-min model prediction. The brushless motor swap eliminated RTD noise from the original design. The box mount redesign made the subsystem CNC-machinable with standard tooling, reducing fabrication cost and enabling in-house production.

MATERIAL SELECTIONS

FLUID CONTACT PARTS **316L SS** Corrosion resistant, biocompatible	VESSEL **Borosilicate** Thermal stability, visibility
INSULATION **Silicone Pad** High-temp, flexible seal	SEALING **O-Ring** IPX-5 rated, static lid

SENSORS AND ELECTRONICS

- **PT100 RTD:** precision temperature, noise-immune
- **MQ-4 methane:** fermentation gas safety monitoring
- **Differential pressure sensor:** ±2 kPa, vessel integrity
- **pH probe:** real-time fermentation tracking
- **ESP32:** micro-SD logging and IoT cloud connectivity

RESULTS SUMMARY

SETUP TIME	THROUGHPUT	TEMP ACCURACY	SPEED CONTROL
79 min	**+25%**	**±0.1°C**	**±1.5 RPM**
From multiple days	Experiments/week	All 5 setpoints	Proportional

Engine Gasket Nonlinear FEA

Hyperelastic Calibration and Shape Optimization in Abaqus

PROJECT OVERVIEW

This project simulated the closure of a rubber engine gasket in Abaqus with the objective of eliminating a sealing gap and producing uniform contact pressure. The workflow covered hyperelastic material calibration from test data, nonlinear FEA with contact mechanics, and iterative shape optimization.

Abaqus/Standard | MATLAB (lsqnonlin) | Ogden N=2 | Contact Mechanics | CPE4RH Elements | Mesh Convergence | Shape Optimization

MATERIAL CALIBRATION

Experimental stress-strain data provided for three deformation modes: simple tension, planar shear, and biaxial tension. Parameters fitted for two hyperelastic constitutive models.

Mooney-Rivlin (2-parameter)

- Nominal stress equations derived analytically for all three deformation modes
- Linear least-squares regression solved in closed form for A10 and A01
- $R^2 = 0.9105$ | RMSE = 0.9418 | SSE = 6.208

Ogden N=2 (4-parameter, Selected)

- Nonlinear optimization via MATLAB lsqnonlin
- $R^2 = 0.9629$ | RMSE = 0.8782 | SSE = 6.171
- 4 parameters allow independent stiffness tuning across stretch ratios

WHY OGDEN OUTPERFORMS MOONEY-RIVLIN

The 2-parameter Mooney-Rivlin model diverges at higher stretch ratios. Ogden's 4 parameters independently tune stiffening across the full stretch range, tracking experimental data closely. The R^2 improvement from 0.9105 to 0.9629 reflects this directly.

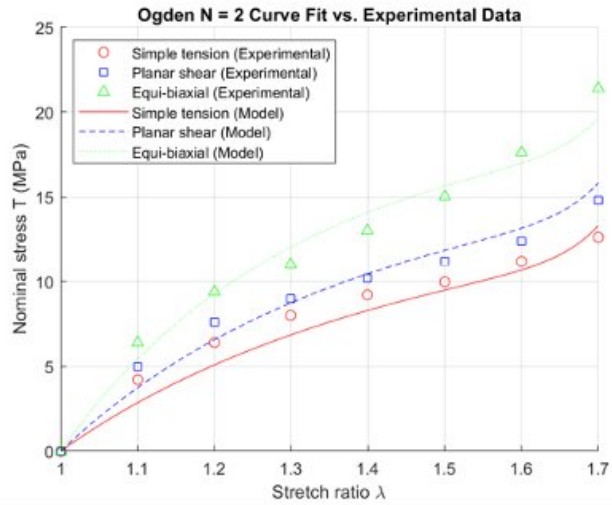

Fig. 10 Ogden N=2 curve fit vs. experimental stress-strain data. $R^2 = 0.9629$.

Model	R^2	RMSE	SSE
Mooney-Rivlin	0.9105	0.9418	6.208
Ogden N=2	**0.9629**	**0.8782**	**6.171**

FEA MODEL SETUP

Element and Mesh

- CPE4RH hybrid elements prevent volumetric locking (rubber, $\nu = 0.498$)
- Quadrilateral elements for convergence under large deformation
- 0.55% change coarse to medium; 0.21% medium to fine — medium mesh (0.7) selected

Contact and BCs

- Hard normal contact, penalty friction $\mu = 0.1$ at all interfaces
- Three contact pairs: head-gasket, block-gasket, gasket self-contact
- Block head compressed 12.6 mm; auto stabilization damping 0.0001

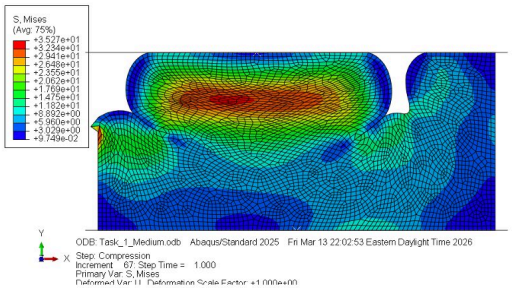

Fig. 11 Initial design, von Mises stress. Stress concentrated at bulb crown. Gap d = 2.2246 mm.

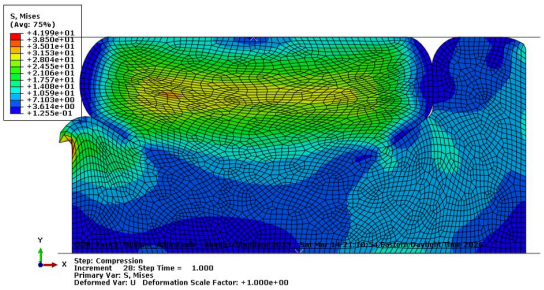

Fig. 12 Optimized design, von Mises stress. Stress redistributed evenly. Gap d = 0 mm.

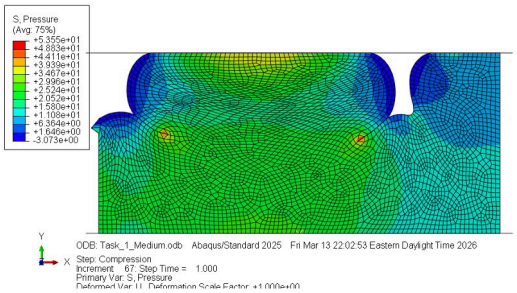

Fig. 13 Initial design, hydrostatic pressure. Contact at narrow center strip only — poor seal distribution.

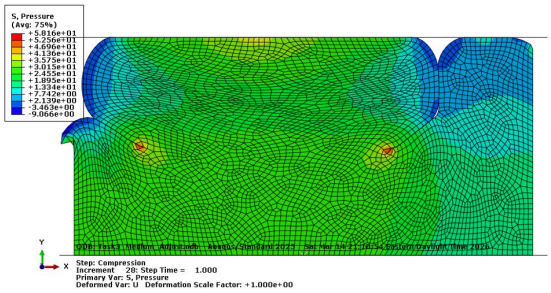

Fig. 14 Optimized design, hydrostatic pressure. Uniform pressure across full contact surface.

INITIAL DESIGN ANALYSIS

The initial geometry produced a gap of d = 2.2246 mm at full compression. Von Mises stress concentrated at the 12.6 mm radius bulb crown, and hydrostatic pressure showed contact force only at a narrow central strip — this configuration would allow engine oil leakage under operating pressure.

FAILURE MODE IDENTIFIED

Small-radius crown concentrates contact force without surface spreading. Under 12.6 mm compression the gasket deforms laterally without closing the gap, leaving a 2.22 mm leakage path.

SHAPE OPTIMIZATION

- Crown: 12.6 mm semicircle changed to 25 mm radius arc, increasing contact width
- Corner fillets: 4 mm radius added at sharp bulb corners
- Base fillets reduced: 3.2 mm to 2 mm, allowing gasket to deform and close gap

Initial	Optimized
Gap d = 2.2246 mm	Gap d = 0 mm
Stress at crown	Stress distributed evenly
Narrow contact strip	Uniform contact pressure

INITIAL GAP	OPTIMIZED GAP	FORCE	R^2 FIT
2.2246 mm	**0 mm**	**+200 N** Acceptable	**0.9629**

No-Slip Walker

Active Fall-Prevention Walker for Elderly Users

ME Design 2, EML 4501 | 8-Member Team
Summer 2024

Fig. 15 Isometric CAD render. 6061-T6 Al frame, 300 lb rated, 147-185 cm height, TPR handles, Arduino Nano.

PROJECT OVERVIEW

The No-Slip Walker is an active fall-prevention mobility aid for elderly users over 60, developed through the double diamond design process. An Arduino Nano controller activates a ratchet-based braking mechanism when unexpected forward lean is detected, locking the rear wheels and preventing fall events.

ANSYS R22.2 | Fusion 360 | GD&T (ASME Y14.5) | 6061-T6 Al | Combined Loading | Arduino Nano

MY CONTRIBUTIONS

- Defined load cases for a 300 lb user in standing and seated positions
- Selected 6061-T6 aluminum: optimal strength-to-weight and machinability
- Ran ANSYS R22.2 structural FEA on 4 components: rear wheel mount, upper frame, lower frame, ratchet
- Identified upper frame as critical (FOS = 1.00) and flagged for redesign
- Confirmed FEA with combined loading hand calc: max allowable 294.7 lbf
- Improved frame stability 20%; produced GD&T drawings per ASME Y14.5

STRUCTURAL FEA RESULTS (ANSYS R22.2)

Component	FOS (Handle)	FOS (Seat)
Rear Wheel Mount	18.54	28.23
Lower Frame	7.45	7.60
Seat Arm	1.72	1.73
Ratchet Mechanism	1.73	1.73
Upper Frame (Critical)	**1.00**	--

CRITICAL FINDING: UPPER FRAME

FOS = 1.00 at worst-case handle loading indicates a failure condition. Combined loading hand calculation (axial + bending) confirmed FEA: max allowable force 294.7 lbf. Flagged for geometry redesign in future iterations. The seat arm (FOS 1.72) and ratchet mechanism (FOS 1.73) are borderline acceptable and should also be monitored in any future redesign.

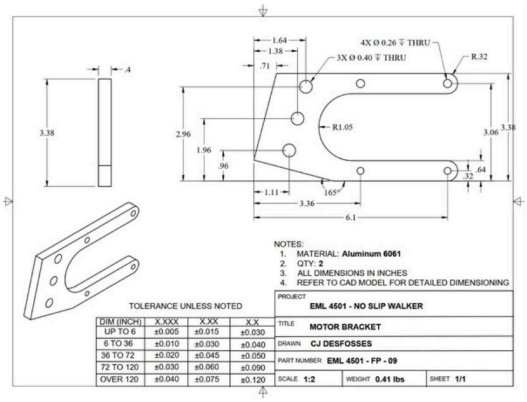

Fig. 16 GD&T detail drawing, motor bracket (EML 4501-FP-09). ASME Y14.5. 6061 Al, 1:2 scale, 0.41 lb.

FEA CONTOUR PLOTS (ANSYS R22.2) — HANDLE LOAD (B) VS. SEAT LOAD (C)

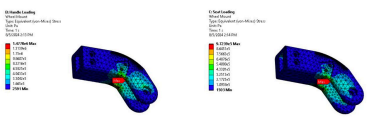

Fig. 17 Rear wheel mount. FOS 18.54 to 28.23. Both load cases well within range.

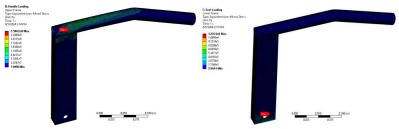

Fig. 18 Upper frame arm: handle load (B) FOS = 1.00 — failure condition, Max stress at joint. Seat load (C) shown right.

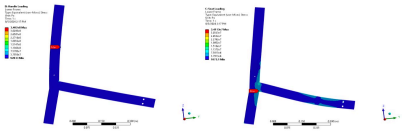

Fig. 19 Lower frame. FOS 7.45 (handle) and 7.60 (seat). Both within acceptable range.